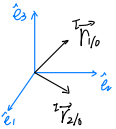
- Generalized Coordinates



N particles described by n generalized coordinates

$\vec{r}_{i/0}(q_1, \cdots q_n) \quad \forall\ i=1,\cdots,N$

Apply newton's law, we get equation of motion in the form of 2nd-order ODE

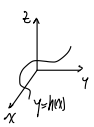
$$\begin{bmatrix} \ddot{q}_1 \\ \vdots \\ \ddot{q}_n \end{bmatrix} = \begin{bmatrix} \text{depends on } q,\ \dot{q} \end{bmatrix}$$

- Constraints and DoF, Holonomic Constraints
  - for a system of N particles in 3 dimension and k constraints
    the DoF is $3N-k$ (for holonomic constraints)
  - a holonomic constraint is a scalar equation that relates generalized coordinates

    $f_k(\vec{r}_{1/0} \cdots \vec{r}_{N/0}, t) = 0 \quad \forall\ k=1,\cdots,K$

    $\phi_k(q_1 \cdots q_n, t) = 0$

    Eg. particle constrained to a curve $y=h(x)$ in xy plane



$f_1(\vec{r}_{1/0}) = z = 0$

$f_2(\vec{r}_{1/0}) = h(x) - y = 0$

$3N-k = 3 \to = 1$ DoF

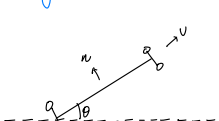
  - For a system with only holonomic constraints, it's always possible to find the minimum set of generalized coordinates (# of generalized coordinate = DoF)

Eg.  9 DoF

  - Variables that describes position and/or orientation are called "configuration variables"

- Non-holonomic Constraints
  - most important for us: Constraints on $\dot{q}$ (generalized velocities)
  - usually linear in $\dot{q}$: $A(q)\dot{q} = 0 \quad A(q) \in \mathbb{R}^{sn}$ s = # of non-holonomic constraints

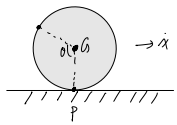
Eg. Shopping Cart



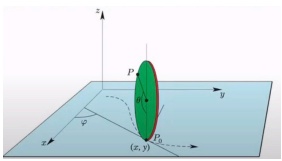
$n^T v = 0 \Rightarrow \begin{bmatrix} -\sin\theta & \cos\theta & 0 \end{bmatrix} \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix} = 0$

$A(q) \qquad \dot{q} = 0$

Eg. Rolling without slipping



$\dot{x} = r\dot{\theta}$
$\dot{x} = r\dot{\theta}$
} Can be written as holonomic or non-holonomic constraints

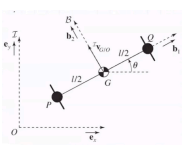


$\dot{x} = \cos\varphi \; r\dot{\theta}$
$\dot{y} = \sin\varphi \; r\dot{\theta}$
} non-holonomic constraints

· System with Holonomic and Non-Holonomic Constraints

$$DoF = \underbrace{3N}_{3 \times \#\text{ particles}} - \underbrace{k}_{\#\text{ holonomic}} - \underbrace{s}_{\#\text{ non-holonomic}}$$

Eg. Ski



$\vec{V}_{a/o}$ and $\vec{V}_{p/o}$ only in $\hat{b}_2$ direction

holonomic constraints: $\| \vec{r}_{p/o} - {}^{I}\vec{r}_{p/o} \|_2 = l$

non-holonomic constraints: $\langle \vec{V}_{a/o}, \hat{b}_{1/2} \rangle = 0$

DoF = 4-1-1 =>
2DoF system, but must be described by 3 generalized coordinates $q = \begin{bmatrix} x_a \\ y_a \\ \theta \end{bmatrix}$ $\begin{bmatrix} \dot{x}_a \\ \dot{y}_a \\ \dot{\theta} \end{bmatrix}^T \begin{bmatrix} -\sin\theta \\ \cos\theta \\ 0 \end{bmatrix} = 0$

DoF is an intrinsic property of the system, independent of the choice of generalized coordinates

if # of generalized coordinates $\left\{ \begin{matrix} = \\ > \end{matrix} \right\}$ # DoF, call it $\left\{ \begin{matrix} \text{minimal} \\ \text{redundant} \end{matrix} \right\}$ set of coordinates

if there's no non-holonomic constraints, it's always theoretically possible to find minimal set of coordinates